Low vegetable intake is associated with allergic asthma and moderate-to-severe airway hyperresponsiveness.

BACKGROUND: In recent decades, children's diet quality has changed and asthma prevalence has increased, although it remains unclear if these events are associated. OBJECTIVE: To examine children's total and component diet quality and asthma and airway hyperresponsiveness (AHR), a proxy for asthma severity. METHODS: Food frequency questionnaires adapted from the Nurses' Health Study and supplemented with foods whose nutrients which have garnered interest of late in relation to asthma were administered. From these data, diet quality scores (total and component), based on the Youth Healthy Eating Index (YHEI adapted) were developed. Asthma assessments were performed by pediatric allergists and classified by atopic status: Allergic asthma (≥1 positive skin prick test to common allergens >3 mm compared to negative control) versus non-allergic asthma (negative skin prick test). AHR was assessed via the Cockcroft technique. Participants included 270 boys (30% with asthma) and 206 girls (33% with asthma) involved in the 1995 Manitoba Prospective Cohort Study nested case-control study. Logistic regression was used to examine associations between diet quality and asthma, and multinomial logistic regression was used to examine associations between diet quality and AHR. RESULTS: Four hundred seventy six children (56.7% boys) were seen at 12.6 ± 0.5 years. Asthma and AHR prevalence were 26.2 and 53.8%, respectively. In fully adjusted models, high vegetable intake was protective against allergic asthma (OR 0.49; 95% CI 0.29-0.84; P < 0.009) and moderate/severe AHR (OR 0.58; 0.37-0.91; P < 0.019). CONCLUSIONS: Vegetable intake is inversely associated with allergic asthma and moderate/severe AHR.